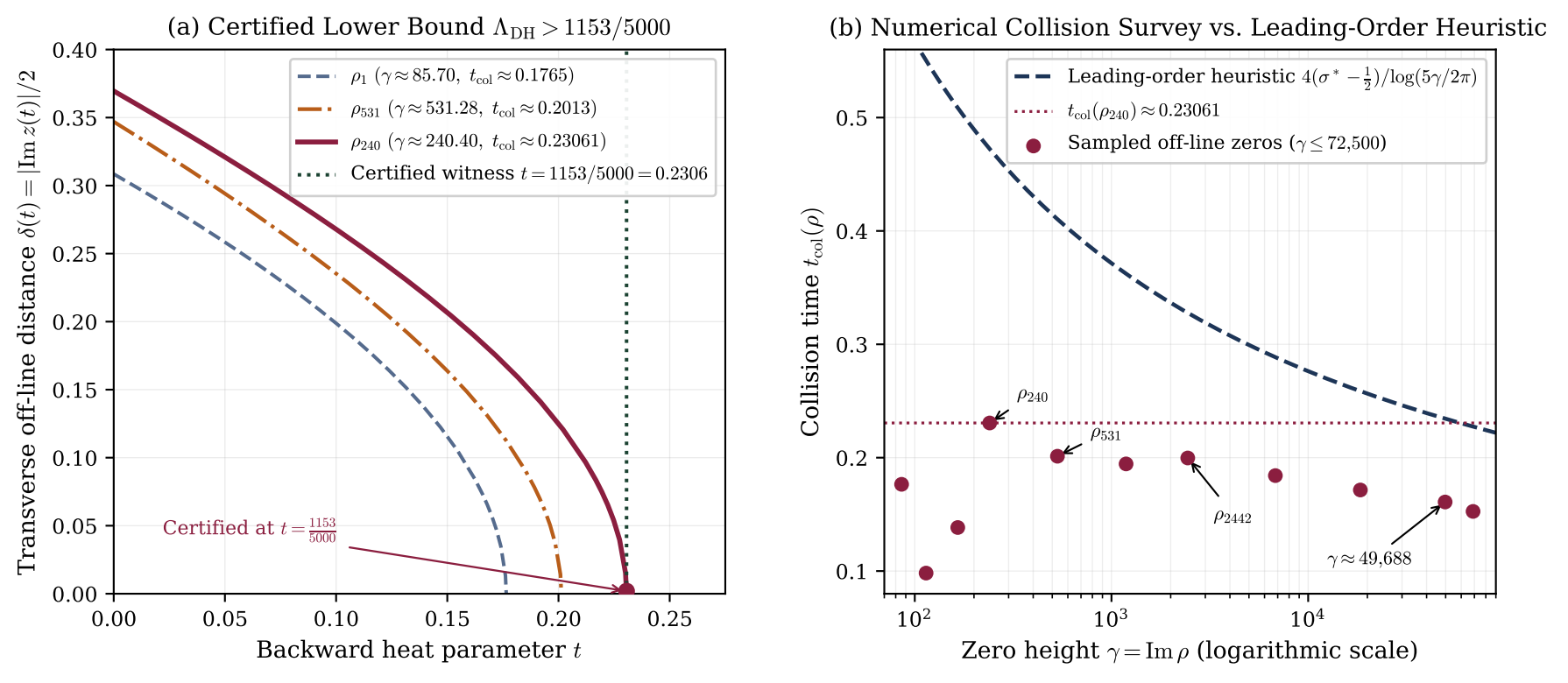
(a) Certified Lower Bound $\Lambda_{\mathrm{DH}} > 1153/5000$
ρ_1 ($\gamma \approx 85.70$, $t_{\mathrm{col}} \approx 0.1765$)
ρ_{531} ($\gamma \approx 531.28$, $t_{\mathrm{col}} \approx 0.2013$)
ρ_{240} ($\gamma \approx 240.40$, $t_{\mathrm{col}} \approx 0.23061$)
Certified witness $t = 1153/5000 = 0.2306$
Certified at $t = \frac{1153}{5000}$
Transverse off-line distance $\delta(t) = |\mathrm{Im}\, z(t)|/2$
Backward heat parameter t
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
(b) Numerical Collision Survey vs. Leading-Order Heuristic
Leading-order heuristic $4(\sigma^* - \frac{1}{2})/\log(5\gamma/2\pi)$
$t_{\mathrm{col}}(\rho_{240}) \approx 0.23061$
Sampled off-line zeros ($\gamma \le 72{,}500$)
ρ_{240}
ρ_{531}
ρ_{2442}
$\gamma \approx 49{,}688$
Collision time $t_{\mathrm{col}}(\rho)$
Zero height $\gamma = \mathrm{Im}\, \rho$ (logarithmic scale)
10^2
10^3
10^4
0.1
0.2
0.3
0.4
0.5